



2025 RTP Economic Analysis Results

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1. 2025 RTP Economic Project Evaluation Results

This section provides details of the evaluation of the selected transmission project improvements using the production cost savings (PCS) test and the congestion cost savings (CCS) test, which is measured by the system-wide consumer energy cost reduction. The analysis was based on the 2027 and 2030 sensitivity cases developed for the economic analysis of the 2025 RTP.

1.1 Elmato POI to Houdini to WA Parish 345-kV Line Upgrade (Project 1)

The Elmato POI to Houdini to WA Parish 345-kV line was congested under the loss of the WA Parish to Wolf and WA Parish to Whaley 345-kV lines. Table 1 includes details of the congestion.

Table 1: Houdini to WA Parish 345-kV Line and Elmato POI to Houdini 345-kV Line Congestion Details

Congested Element	Houdini to WA Parish 345-kV line			
Most Severe Contingency	Loss of the WA Parish to Wolf and WA Parish to Whaley 345-kV lines			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1325	2027	24.33	150.3
		2030	6.69	22.0
Congested Element	Elmato POI to Houdini 345-kV line			
Most Severe Contingency	Loss of the WA Parish to Wolf and WA Parish to Whaley 345-kV lines			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1325	2027	24.33	32.6
		2030	6.69	14.0

Upgrading the Elmato POI to Houdini to WA Parish 345-kV line alleviated the congestion in the area without creating any additional congestion. The project was evaluated in the 2030 economic case, and the economic analysis results are included in Table 2. The project demonstrated an annual decrease of \$3.9 million in production cost and the break-even capital cost is \$29.9 million. On the other hand, the project caused an increase of \$20.9 million in the annual consumer energy cost.

Table 2: Economic Analysis Results for the Elmato POI to Houdini to WA Parish 345-kV Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	-	4.1	3.9	29.9
CCS Test	-	-22.3	-20.9	-

1.2 Friendswood to Webster and Friendswood to Mustang Bayou 138-kV Line Upgrades (Project 2)

The Friendswood to Webster 138-KV line was congested under the loss of the Meadow to Oasis 345-kV line. Table 3 includes details of the congestion on this line. To resolve the congestion, the following lines were upgraded: the Friendswood to Webster 138-KV line and the Friendswood to Mustang Bayou 138-kV line.

Table 3: Friendswood – Webster 138-kV Line Congestion Details

Congested Element	Friendswood to Webster 138-kV line			
Most Severe Contingency	Loss of the Meadow to Oasis 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	561	2027	-	-
		2030	13.94	17

The project was evaluated in the 2030 economic case, and the economic analysis results are included in Table 4. The project demonstrated an annual decrease of \$0.9 million in production cost and the break-even capital cost is \$6.7 million. Moreover, the project resulted in a decrease of \$7.3 million in the annual consumer energy cost and the break-even capital cost is \$57.5 million.

Table 4: Economic Analysis Results for the Friendswood to Webster and Friendswood to Mustang Bayou 138-kV Line Upgrades

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	-	0.9	0.9	6.7
CCS Test	-	7.8	7.3	57.5

1.3 Fisher Road Switch to Short Creek Solar POI 345-kV Line and Fisher Road Switch to Wichita Falls 138-kV Line Upgrade (Project 3)

The Fisher Road Switch to Short Creek Solar POI 345-kV line was congested under the loss of Bowman Switch to Byrd Road Switch 345-kV line and the Bowman Switch to Riley 345-kV line. Table 5 includes details of the congestion on this line.

Table 5: Fisher Road Switch to Short Creek Solar POI 345-kV Line Congestion Details

Congested Element	Fisher Road Switch to Short Creek Solar POI 345-kV line			
Most Severe Contingency	Loss of the Bowman Switch to Byrd Road Switch 345-kV line and the Bowman Switch to Riley 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1251	2027	-	-
		2030	15.96	30

Upgrading the Fisher Road Switch to Short Creek Solar POI 345-kV line and the Fisher Road Switch to Wichita Falls 138-kV line alleviated the congestion in the area without creating any additional congestion.

The project was evaluated in the 2030 economic case, and the economic analysis results are included in Table 6. The project demonstrated an annual decrease of \$1.0 million in production cost and the break-even capital cost is \$7.8 million. On the other hand, the project reduced the annual consumer energy cost by \$24.7 million, and the break-even capital cost is \$194.3 million.

Table 6: Economic Analysis Results for the Fisher Road Switch to Short Creek Solar POI 345-kV Line and Fisher Road Switch to Wichita Falls 138-kV Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	-	1.1	1.0	7.8
CCS Test	-	26.3	24.7	194.3

1.4 Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV Line Upgrade (Project 4)

The Buntin Drive to Simpson Stuart 138-kV line was congested in 2030 under the loss of the West Levee Switch to Green Road 345-kV line and the Sargent Road to Green Road 345-kV line. Table 7 includes the details of the congestion on this line.

Table 7: Buntin Drive to Simpson Stuart 138-kV Line Congestion Details

Congested Element	Buntin Drive to Simpson Stuart 138-kV line			
Most Severe Contingency	Loss of the West Levee Switch to Green Road 345-kV line and the Sargent Road to Green Road 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	614	2027	-	-
	245 ¹	2030	4.75	37.4

Upgrading the Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line alleviated the congestion in the area without creating any additional congestion.

The project was evaluated in the 2030 economic case, and the economic analysis results are included in Table 8. The project demonstrated an annual decrease of \$1.3 million in production cost and the break-even capital cost is \$9.8 million. Additionally, the project demonstrated an annual decrease of \$68.6 million in consumer energy cost and the break-even capital cost is \$540.3 million. Though the 2024 RTP did not identify the reliability need for this upgrade once the 765-kV Strategic Transmission Expansion Plan (STEP) infrastructure is in place, the economic analysis showed sufficient economic benefits based on the generic cost estimates.

Table 8: Economic Analysis Results for the Simpson Stuart to Buntin Drive to Witt Road to Green Road 138-kV Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	-	1.4	1.3	9.8
CCS Test	-	73.2	68.6	540.3

¹ The 2030 case inherited the transmission networks from the 2024 RTP 2030 final reliability case with 765-kV transmission networks. The case did not include the reliability project, 2024-NC73, which was included in the 2024 RTP 2027 final reliability case.

1.5 Bell County East to Voss Lake 345-kV Double-Circuit Line Upgrade (Project 5)

The Bell County East to Voss Lake 345-kV double-circuit line was congested in 2027 under the loss of Hutto Switch to Salado Switch 345-kV double-circuit line. This congestion did not appear in the 2030 case because the start case included a 2024 RTP reliability project (2024-SC28) that upgraded this line. Table 9 includes details of the congestion on this line.

Table 9: Bell County East to Voss Lake 345-kV Double-Circuit Line Congestion Details

Congested Element	Bell County East to Voss Lake 345-kV double-circuit line			
Most Severe Contingency	Loss of the Hutto Switch to Salado Switch 345-kV double circuit line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1072	2027	30.97	111.7
	1912	2030	-	-

While the 2024 RTP reliability analysis found that the Bell County East to Voss Lake 345-kV double-circuit line upgrade could be needed starting in 2029, the 2025 RTP reliability analysis results showed that this project may be needed as early as in 2027. The project was evaluated in the 2027 economic case and the economic analysis results are included in Table 10. The project demonstrated an annual decrease of \$40.5 million in production cost. Additionally, the project demonstrated an annual decrease of \$29.4 million in consumer energy cost. The results indicated that the project could produce significant economic savings besides its reliability benefits.

Table 10: Economic Analysis Results for the Bell County East to Voss Lake 345-kV Double-Circuit Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	40.5	-	-	-
CCS Test	29.4	-	-	-

1.6 Four Brothers Switch to Tradinghouse SES to Lake Creek 345-kV Line Upgrade (Project 6)

The Four Brothers Switch to Tradinghouse SES 345-kV line was congested under the loss of the Venus Switch to Fort Smith Switch 345-kV and Miller Road to Forbes Solar 345-kV lines. Table 11 includes details of the congestion on this line.

Table 11: Four Brothers Switch to Tradinghouse SES 345-kV Line Congestion Details

Congested Element	Four Brothers Switch to Tradinghouse SES 345-kV line			
Most Severe Contingency	Loss of the Venus Switch to Fort Smith Switch 345-kV and Miller Road to Forbes Solar 345-kV lines			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1912	2027	-	-
	1072 ²	2030	18.64	37.4

² The 2030 case inherited the transmission networks from the 2024 RTP 2030 reliability case with 765-kV transmission networks. The case did not include the reliability project, 2024-NC32, which was included in the 2024 RTP 2027 final reliability case.

Upgrading the Four Brothers Switch to Tradinghouse SES to Lake Creek 345kV line alleviated the congestion in the area without creating any additional congestion.

The project was evaluated in the 2030 economic case, and the economic analysis results are included in Table 12. The project demonstrated an annual decrease of \$4.9 million in production cost and the break-even capital cost is \$37.9 million. Additionally, the project demonstrated an annual decrease of \$3.5 million in consumer energy cost and the break-even capital cost is \$27.7 million.

Table 12: Economic Analysis Results for the Four Brothers Switch to Tradinghouse SES to Lake Creek 345kV Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	-	5.3	4.9	37.9
CCS Test	-	3.8	3.5	27.7

1.7 Mackenzie Substation to Northeast Substation to Dunbar Substation 115-kV Line Upgrade (Project 7)

The Mackenzie Substation to Northeast Substation 115-kV line was congested in 2027 under the loss of the Blackwater Draw Switch to Double Mountain Switch 345-kV line. This congestion did not appear in the 2030 case because the start case included a 2024 RTP reliability project (2024-N08) that upgraded this line. Table 13 includes details of the congestion on this line.

Table 13: Mackenzie Substation to Northeast Substation 115-kV Line Congestion Details

Congested Element	Mackenzie Substation to Northeast Substation 115-kV line			
Most Severe Contingency	Loss of the Blackwater Draw Switch to Double Mountain Switch 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	218	2027	14.82	31.2
	431	2030	-	-

Both the 2024 RTP and 2025 RTP reliability analysis identified the need to upgrade the Mackenzie Substation to Northeast Substation 115-kV line (2024-N08 and 2025-N08). While the 2024 RTP reliability analysis showed that the project was needed starting in 2029, the 2025 RTP reliability analysis identified the need to be as early as 2028.

The project was evaluated in the 2027 economic case, and it was able to alleviate the congestion in the area without creating any additional congestion. The economic analysis results are included in Table 14. The project demonstrated an annual decrease of \$34.1 million in production cost. On the other hand, the project caused an increase of \$12.7 million in the annual consumer energy cost. The results indicated that the project may produce significant production cost savings besides its reliability benefits.

Table 14: Economic Analysis Results for the Mackenzie Substation to Northeast Substation to Dunbar Substation 115-kV Line Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	34.1	-		-
CCS Test	-12.7	-		-

1.8 Carver to Poblano 69-kV to 138-kV Line Conversion (Project 8)

The Barksdale to Campwood 69-kV line was congested in 2027 under the loss of the Kendall to Edison 345-kV double-circuit line. Table 15 includes details of the congestion on this line.

Table 15: Barksdale to Campwood 69-kV Line Congestion Details

Congested Element	Barksdale to Campwood 69-kV line			
Most Severe Contingency	Loss of the Kendall to Edison 345-kV double-circuit line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	38	2027	6.54	16
		2030	-	-

The following planned Tier 4 project, which is projected to be in-service in December 2028, was evaluated in the 2027 economic case. The 2025 RTP reliability analysis also showed that the need for this Tier 4 project (2025-W03) could be as early as in 2027.

- Convert Carver to Bondroad to Friess Ranch to Rocksprings to New Barksdale to Campwood to Montell to Poblano to Uvalde lines from 69-kV to 138-kV such that their new emergency ratings are at least 485 MVA.

The economic analysis results showed that this project was able to alleviate congestion in the area without creating any additional congestion. The economic analysis results are included in Table 16. The project demonstrated an annual decrease of \$28.9 million in production cost. On the other hand, the project caused an increase of \$10.8 million in the annual consumer energy cost. It was shown that the project can produce significant production cost savings besides its reliability benefits when accelerated to 2027.

Table 16: Economic Analysis Results for Carver to Poblano 69-kV to 138-kV Line Conversion

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	28.9	-		-
CCS Test	-10.8	-		-

1.9 Sulphur Springs Switch to Liberty to Edgewood 138-kV Line Upgrade (Project 9)

The Sulphur Springs to Liberty 138-kV line was congested under the loss of the Sulphur Springs Switch to Cartwheel POI 345-kV line, the Pineforest POI to Allen Switch 345-kV line, and the Monticello transformer. Table 17 includes details of the congestion on this line.

Table 17: Sulphur Springs to Liberty 138-kV Line Congestion Details

Congested Element	Sulphur Springs to Liberty 138-kV line			
Most Severe Contingency	Loss of the Sulphur Springs Switch to Cartwheel POI 345-kV line, the Pineforest POI to Allen Switch 345-kV line, and the Monticello transformer			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	203	2027	3.45	3.8
		2030	4.69	13.6

Upgrading the 138-kV line from Sulphur Springs to Liberty to Edgewood alleviated the congestion in the area without creating any additional congestion.

The project was evaluated in the 2027 and 2030 economic cases, and the economic analysis results are included in Table 18. The project demonstrated an annual decrease of \$10.4 million in production cost. On the other hand, the project caused an increase of \$2.2 million in the annual consumer energy cost. The study results also showed that the factors driving the economic benefits, due to this project, may not continue in the long run as the production cost savings in 2030 were much less compared to those in 2027. Due to this reason, the levelized savings and the break-even capital cost were not calculated.

Table 18: Economic Analysis Results for the Sulphur Springs to Liberty to Edgewood 138-kV Lines Upgrade

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	19.4	2.8	-	-
CCS Test	-10.7	6.5	-2.2	-

1.10 Fowlerton Area Improvement (Project 10)

The Tilden to Fowlerton 138-kV line was congested under the loss of the Fowlerton to San Miguel 345-kV double-circuit line. Additionally, in the same area, the Bruni Sub 138/69-kV transformer was congested under the loss of the Lobo to Laquinta 138-kV line. Table 19 includes details of the congestion on the Tilden to Fowlerton 138-kV line and the Bruni Sub 138/69-kV transformer.

Table 19: Tilden to Fowlerton 138-kV Line and Bruni Sub 138/69-kV Transformer Congestion Details

Congested Element	Tilden to Fowlerton 138-kV line			
Most Severe Contingency	Loss of the Fowlerton to San Miguel 345-kV Double-Circuit Line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	259	2027	15.00	16.7
		2030	9.62	15.5
Congested Element	Bruni Sub 138/69-kV transformer			
Most Severe Contingency	Loss of the Lobo to Laquinta 138-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	37	2027	26.94	10.8
		2030	16.39	9.1

The following improvements were made in the Fowlerton area to alleviate these congestions in the area without creating any additional congestion:

- Upgrade of the Fowlerton to Tilden to George West 138-kV line, the Bruni 138/69-kV transformer, the Bruni to Freer to Holland to Cotulla 69-kV lines; and
- Addition of new 138-kV lines from George West to Three Rivers and from Fowlerton to Hindes.

The project was evaluated in the 2027 and 2030 economic cases, and the economic analysis results are included in Table 20. The project demonstrated an annual decrease of \$6.4 million in production cost and the break-even capital cost is \$49.2 million. Additionally, the project demonstrated an annual decrease of \$26.6 million in consumer energy cost and the break-even capital cost is \$209.5 million.

Table 20: Economic Analysis Results for Fowlerton Area Improvement

	Adjusted Savings in 2027 (M\$)	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	11.4	2.3	6.4	49.2
CCS Test	22.9	35.9	26.6	209.5

1.11 White River to Long Draw 345-kV Double-Circuit Line Addition (Project 11)

The Panhandle interface remained one of the top congested paths for both the 2027 and 2030 study years. Moreover, one of the key paths delivering energy from Panhandle to West Texas, the Farmland to Long Draw 345-kV line, was heavily congested under the loss of another path, the Buzios Solar to Cottonwood 345-kV double-circuit line, between Panhandle and West Texas. Table 21 includes details of the congestion on this interface and nearby lines.

Table 21: Panhandle Interface, Holly POI to Long Draw 345-kV Line and Farmland to Holly POI 345-kV Line Congestion Details

Congested Element	Panhandle interface			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	4440	2027	13.5	11.9
		2030	18.8	45.5
Congested Element	Holly POI to Long Draw 345-kV line			
Most Severe Contingency	Loss of the Buzios Solar to Cottonwood 345-kV double-circuit line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1031	2027	17.6	9.3
	1072	2030	29.7	46.3
Congested Element	Farmland to Holly POI 345-kV line			
Most Severe Contingency	Loss of the Buzios Solar to Cottonwood 345-kV double-circuit line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1031	2027	15.7	34.9
	1072	2030	17.5	8.2

To address the above-mentioned issue, a project adding a new 345-kV double-circuit line to connect White River to Long Draw was tested in the 2030 economic cases. This project increased the transfer limit on the Panhandle interface and in turn reduced congestion on it. It also served as an additional path delivering energy from Panhandle to Far West and reduced the congestion on the Farmland to Long Draw 345-kV line.

The economic analysis results are included in Table 22. The project reduced congestion rent on Panhandle interface, the Holly POI to Long Draw 345-kV line, and the Farmland to Holly POI 345-kV line by \$38.4 million, \$43 million and \$11.8 million respectively. However, it added congestion to the West Texas Export (WTX) interface since part of the additional energy coming from the Panhandle passes through the WTX interface to reach the demand centers.

The project demonstrated an annual decrease of \$28.2 million in production cost and the break-even capital cost is \$203.2 million. Additionally, the project showed an annual decrease of \$26.6 million in consumer energy cost and the break-even capital cost is \$196 million. Overall, the project reduced the system-wide congestion rent by \$81 million.

Table 22: Economic Analysis Results for the White River to Long Draw 345-kV Double-Circuit Line Addition

	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	28.2	26.4	203.2
CCS Test	26.6	24.9	196

1.12 North to Houston Import Improvement Projects (Project 12)

The North to Houston interface congestion rent was observed at \$47.9 million for 2027 and \$140.6 million for 2030 in sensitivity cases. Table 23 includes details of the congestion on this interface.

To test different projects impact on this congestion, VSAT analysis was performed on various options to improve the North to Houston interface limit and decrease the congestion on this path. Options listed below were the ones with better economic potential.

Table 23: North to Houston Interface Congestion Details

Congested Element	North to Houston interface			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	5595 ³	2027	11.8	47.9
		2030	25.6	140.6

1.12.1 North to Houston Import Improvement Option a

This option includes a new 345-kV double-circuit line from Tomball to Zenith to Hillje and the upgrade of the existing Zenith to TH Wharton 345-kV line. It increased the North to Houston interface limit and thus helped with the delivery of more energy to the greater Houston area. The project was evaluated for the 2030 study year. The economic analysis results are included in Table 24. This project decreased the congestion rent on the North to Houston interface by \$50.5 million for the 2030 study year. The project demonstrated an annual decrease of \$16.1 million in production cost and the break-even capital cost is \$115.9 million. Additionally, the project demonstrated an annual increase of \$82.1 million in consumer energy cost while reducing the system-wide congestion rent by \$44 million.

Table 24: Economic Analysis Results for the North to Houston Import Improvement (Option a)

	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	16.1	15.1	115.9
CCS Test	-82.1	-77	-

1.12.2 North to Houston Import Improvement Option b

To support the voltage under the critical contingency limiting the North to Houston interface transfer capability, option a was further refined by adding two 100 MVar capacitor banks to form Option b. The economic analysis results are included in Table 25. This project decreased the congestion rent on the North to Houston interface by \$54 million for the 2030 study year. It also demonstrated an annual decrease of \$18.7 million in production cost and the break-even capital cost is \$134.9 million. Additionally, the project demonstrated an annual increase of \$93.8 million in consumer energy cost while reducing the system-wide congestion rent by \$45.9 million.

³ Note that North to Houston interface limit in the economic analysis (UPLAN) was modeled with hourly profiles. Details can be found at https://www.ercot.com/files/docs/2025/08/05/2025-Regional-Transmission-Plan-RTP-Economic-Study-Stability-Interface-Limits-Update-8_04_25-.pdf

Table 25: Economic Analysis Results for the North to Houston Import Improvement (Option b)

	Adjusted Savings in 2030 (M\$)	Levelized Savings in 2027 Dollars (\$M)	Break-even Capital Cost in 2027 Dollars (\$M)
PCS Test	18.7	17.5	134.9
CCS Test	-93.8	-87.9	-

2. Economic Analysis Result Summary

The economic benefits for 2025 RTP economic projects were calculated for the PCS test, and CCS test, as shown in Table 26. The break-even capital costs for 2025 RTP economic projects, if applicable, are provided in Table 27. All cost savings and the break-even capital cost listed here are in 2027 dollars. The key findings are as follows:

- Based on the generic cost estimates, the following projects can potentially meet the economic planning criteria. However, the ultimate determination of the economic viability of the proposed transmission projects should be based on detailed cost estimates provided by the TSPs.
 - Friendswood to Webster and Friendswood to Mustang Bayou 138-kV line upgrades (projects 2)
 - Fisher Road Switch to Short Creek Solar POI 345-kV line and Fisher Road Switch to Wichita Falls 138-kV line upgrade (project 3)
 - Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line upgrade (project 4)
- Accelerating the in-service date of the Tier 4 Carver to Poblano 69-kV to 138-kV line conversion project (project 8) to 2027 provided significant economic benefits.

Table 26: Economic Analysis Results for the 2025 RTP Economic Project Evaluation

Index	Description	Levelized Production Cost Savings (\$M)	Levelized Consumer Energy Cost Reduction (\$M)
Project 1	Elmato POI to Houdini to WA Parish 345-kV line upgrade	3.9	-20.9
Project 2	Friendswood to Webster and Friendswood to Mustang Bayou 138-kV line upgrade	0.9	7.3
Project 3	Fisher Road Switch to Short Creek Solar POI 345-kV line and Fisher Road Switch to Wichita Falls 138-kV line upgrade	1.0	24.7
Project 4	Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line upgrade	1.3	68.6
Project 5	Bell County East to Voss Lake 345-kV double-circuit line upgrade	-	-
Project 6	Four Brothers Switch to Tradinghouse SES to Lake Creek 345-kV line upgrade	4.9	3.5
Project 7	Mackenzie Substation to Northeast Substation to Dunbar Substation 115-kV line upgrade	-	-
Project 8	Carver to Poblano 69-kV to 138-kV line conversion	-	-
Project 9	Sulphur Springs Switch to Liberty to Edgewood 138-kV line upgrade	-	-2.2
Project 11	Fowlerton area improvement ⁴	6.4	26.6
Project 11	White River to Long Draw 345-kV double-circuit line addition	26.4	24.9
Project 12. a	North to Houston import improvement option a ⁵	15.1	-77
Project 12. b	North to Houston import improvement option b ⁶	17.5	-87.9

⁴ Upgrade of the Fowlerton to Tilden to George West 138-kV line, the Bruni 138/69-kV transformer, the Bruni to Freer to Holland to Cotulla 69-kV line and addition of new 138-kV lines from George West to Three Rivers and from Fowlerton to Hindes

⁵ Addition of 345-kV double-circuit lines between Zenith to Tomball and Zenith to Hillje as well as upgrading the 345-kV line between Zenith to TH Wharton

⁶ Option a plus 2x100 MVAR capacitor banks

Table 27: Break-even Capital Cost for the 2025 RTP Economic Project Evaluation

Index	Description	PCS Test Break-Even Capital Cost (\$M)	CCS Test Break-Even Capital Cost (\$M)
Project 1	Elmato POI to Houdini to WA Parish 345-kV line upgrade	29.9	-
Project 2	Friendswood to Webster and Friendswood to Mustang Bayou 138-kV line upgrade	6.7	57.5
Project 3	Fisher Road Switch to Short Creek Solar POI 345-kV line and Fisher Road Switch to Wichita Falls 138-kV line upgrade	7.8	194.3
Project 4	Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line upgrade	9.8	540.3
Project 5	Bell County East to Voss Lake 345-kV double-circuit line upgrade	-	-
Project 6	Four Brothers Switch to Tradinghouse SES to Lake Creek 345-kV line upgrade	37.9	27.7
Project 7	Mackenzie Substation to Northeast Substation to Dunbar Substation 115-kV line upgrade	-	-
Project 8	Carver to Poblano 69-kV to 138-kV line conversion	-	-
Project 9	Sulphur Springs Switch to Liberty to Edgewood 138-kV line upgrade	-	-
Project 10	Fowlerton area improvement ⁷	49.2	209.5
Project 11	White River to Long Draw 345-kV double-circuit line addition	203.2	196
Project 12. a	North to Houston import improvement option a ⁸	115.9	-
Project 12. b	North to Houston import improvement option b ⁹	134.9	-

⁷ Upgrade of the Fowlerton to Tilden to George West 138-kV line, the Bruni 138/69-kV transformer, the Bruni to Freer to Holland to Cotulla 69-kV line and addition of new 138-kV lines from George West to Three Rivers and from Fowlerton to Hindes

⁸ Addition of 345-kV double-circuit lines between Zenith to Tomball and Zenith to Hillje as well as upgrading the 345-kV line between Zenith to TH Wharton

⁹ Option a plus 2x100 MVAR capacitor banks